

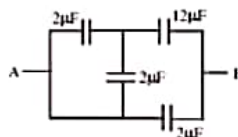
LEVEL-I (MAIN)

EXERCISE-I

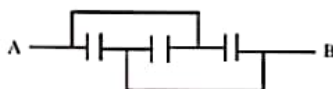
Straight Objective Type Questions

Capacitors and its combination :

1. A body is charged to a certain potential. When, an additional charge of 60 nC is imparted to it, the rise in potential is found to be 15 mV. Find the capacitance of the body.
1) $6\mu\text{F}$ 2) $8\mu\text{F}$ 3) $4\mu\text{F}$ 4) $2\mu\text{F}$
2. Three connected conductors A, B and C have a total charge of $48\mu\text{C}$. The ratio of their capacitances are 1 : 3 : 2. The charges on them individually.
1) $24\mu\text{C}, 12\mu\text{C}, 12\mu\text{C}$ 2) $8\mu\text{C}, 18\mu\text{C}, 22\mu\text{C}$ 3) $8\mu\text{C}, 24\mu\text{C}, 16\mu\text{C}$ 4) $16\mu\text{C}, 16\mu\text{C}, 16\mu\text{C}$
3. The radius of the circular plates of a parallel plate condenser is 'r'. Air is there as the dielectric. The distance between the plates if its capacitance is equal to that of an isolated sphere of radius 'r' is
1) $\frac{r^2}{4r}$ 2) $\frac{r^2}{r}$ 3) $\frac{r}{r}$ 4) $\frac{r^2}{4}$
4. A capacitor acquires a p.d. of 200V when 10^{12} electrons are taken from one plate and placed on the other plate. Its capacitance is
1) $2 \times 10^{-10}\text{F}$ 2) $4 \times 10^{-10}\text{F}$ 3) $8 \times 10^{-10}\text{F}$ 4) $12 \times 10^{-10}\text{F}$
5. The capacity of a parallel plate capacitor of each plate area of $10\text{cm} \times 10\text{cm}$ and separated by a distance 1 mm
1) 88.42 PF 2) 8 PF 3) 2 PF 4) 10 PF
6. The capacities of three capacitors are in ratio 1 : 2 : 3. Their equivalent capacity when connected in parallel is $\frac{60}{11}\mu\text{F}$ more than that when they are connected in series. The individual capacitors are of capacities in μF .
1) 4, 6, 7 2) 1, 2, 3 3) 2, 3, 4 4) 1, 3, 6
7. Three condensers of capacity 4 mF, 2 mF and 3 mF are connected such that 2 mF and 3 mF are in series and 4 mF is parallel to them. The equivalent capacity of the combination is
1) 9 mF 2) 5.2 mF 3) 2.6 mF 4) 10 mF
8. The effective capacitance in μF in between A and B will be
1) 28/9
2) 4
3) 5
4) 18



9. In the given figure each capacitor is equal to $45\mu\text{F}$ then the equivalent capacity between A and B in the given circuit is



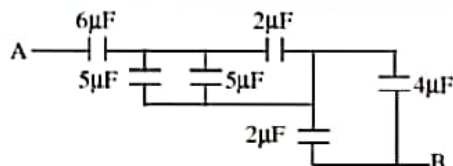
- 1) $15\mu\text{F}$ 2) $10\mu\text{F}$ 3) $40\mu\text{F}$ 4) $135\mu\text{F}$

10. Four plates of the same area of cross section are joined as shown in figure. The distance between each plate is d . The equivalent capacity across A and B will be



- 1) $\frac{2\epsilon_0 A}{d}$ 2) $\frac{3\epsilon_0 A}{d}$ 3) $\frac{3\epsilon_0 A}{2d}$ 4) $\frac{\epsilon_0 A}{d}$

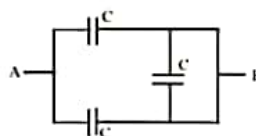
11. The equivalent capacitance between A and B in the circuit given below, is :



- 1) $2.4\mu f$ 2) $4.9\mu f$ 3) $3.6\mu f$ 4) $5.4\mu f$

12. The equivalent capacity of the combination is

- 1) C 2) $2C$
3) $C/2$ 4) $3C$

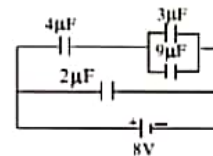


13. Two connected bodies having respective capacitances C_1 and C_2 are charged with a total charge Q . The potentials of the two bodies are,

- 1) $\frac{Q}{C_1+C_2} \cdot \frac{Q}{C_1+C_2}$ 2) $\frac{Q}{C_1} + \frac{Q}{C_2} \cdot \frac{Q}{C_1} + \frac{Q}{C_2}$ 3) $\frac{C_1 C_2}{C_1+C_2} \cdot \frac{C_1 C_2}{C_1-C_2}$ 4) $\frac{Q}{C_1} - \frac{Q}{C_2} \cdot \frac{Q}{C_1-C_2}$

14. A combination of capacitors is set up as shown in the figure. The magnitude of the electric field, due to a point charge Q (having a charge equal to the sum of the charges on the $4\mu F$ and $9\mu F$ capacitors), at a point distant 30m from it, would equal:

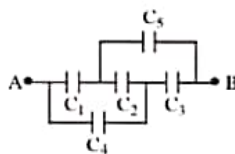
- 1) 480 N/C
2) 240 N/C
3) 360 N/C
4) 420 N/C



15. Assume the earth to be spherical conductor of radius 6400km, the capacitance is

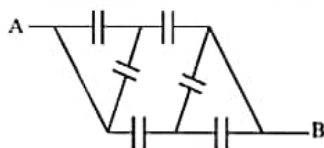
- 1) 711 F 2) $711\mu F$ 3) 710F 4) $71\mu F$

16. The equivalent capacity between A and B in the given circuit is ($C_1 = 4\mu F$, $C_2 = 12\mu F$, $C_3 = 8\mu F$, $C_4 = 4\mu F$, $C_5 = 8\mu F$)



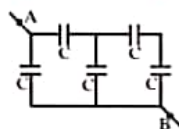
- 1) $24\mu F$ 2) $36\mu F$ 3) $\frac{16}{3}\mu F$ 4) $\frac{8}{3}\mu F$

17. The capacity of each condenser in the following fig. is 'C'. Then the equivalent capacitance across A and B is



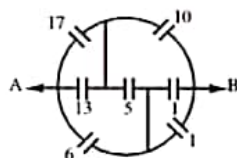
- 1) $C/4$ 2) $3C/4$ 3) $4C/3$ 4) $3C$

18. What is the equivalent capacitance of the system of capacitor between A & B



- 1) $\frac{7}{6}C$ 2) $1.6C$ 3) C 4) None

19. Find equivalent capacitance across AB (all capacitances in μF)



- 1) $\frac{20}{3}\mu\text{F}$ 2) $9\mu\text{F}$ 3) $48\mu\text{F}$ 4) None

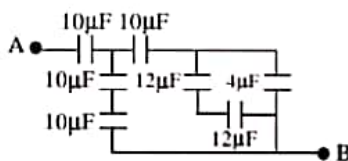
20. A technician has only two capacitors. By using these singly in series or in parallel he is able to obtain the capacitances of $3\mu\text{F}$, $4\mu\text{F}$, $12\mu\text{F}$ and $16\mu\text{F}$. The capacitances of capacitors are

- 1) $6\mu\text{F}$ and $10\mu\text{F}$ 2) $4\mu\text{F}$ and $12\mu\text{F}$ 3) $7\mu\text{F}$ and $9\mu\text{F}$ 4) $4\mu\text{F}$ and $16\mu\text{F}$

21. Three condensers of capacities 3 mF , 6 mF , 12 mF are connected in series with a battery. If the charge on 12 mF condenser is 24 mC , the P.D. across the battery is

- 1) 2 V 2) 4 V 3) 8 V 4) 14 V

22. In the given circuit the effective capacity between A and B is



- 1) $20\mu\text{F}$ 2) $5\mu\text{F}$ 3) $30\mu\text{F}$ 4) $10\mu\text{F}$

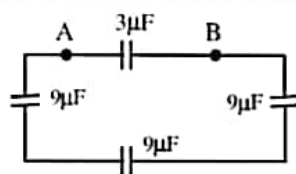
23. The diameter of each plate of an air capacitor is 4 cm . To make the capacity of this plate capacitor equal to that of a sphere of diameter 20 cm , the distance between the plates will be

- 1) $4 \times 10^{-3}\text{ m}$ 2) $1 \times 10^{-3}\text{ m}$ 3) 1 cm 4) $1 \times 10^{-3}\text{ cm}$

24. The capacity of a parallel plate capacitor formed by the plates of same area A is $0.02 \mu\text{F}$ with air as dielectric. Now one plate is replaced by a plate of area $2A$ and dielectric ($K=2$) is introduced between the plates, the capacity is

1) $0.04 \mu\text{F}$ 2) $0.08 \mu\text{F}$ 3) $0.01 \mu\text{F}$ 4) $2 \mu\text{F}$

25. The equivalent capacitance between 'A' and 'B' in the adjoining figure is

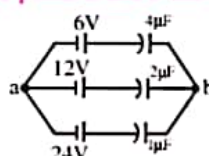


1) $9 \mu\text{F}$ 2) $3 \mu\text{F}$ 3) $6 \mu\text{F}$ 4) $18 \mu\text{F}$

Capacitor circuits :

26. Find the potential difference $V_a - V_b$ between the points a and b shown in each part of the figure.

1) -5.2 V
2) -6.3 V
3) -10.3 V
4) -12.5 V

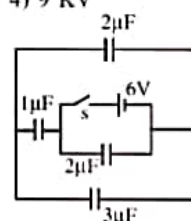


27. A capacitor 1mF withstands a maximum voltage of 6KV while another capacitor 2mF withstands a maximum voltage of 4KV . If the capacitors are connected in series, the system will withstand a maximum voltage of (MNR)

1) 2 KV 2) 4 KV 3) 6 KV 4) 9 KV

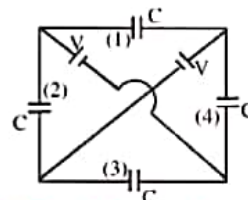
28. Find the charge supplied by the cell after the switch is closed

1) $8 \mu\text{C}$
2) $6 \mu\text{C}$
3) $16.8 \mu\text{C}$
4) $10 \mu\text{C}$

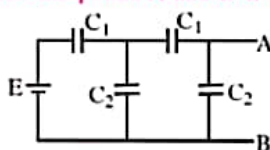


29. In the circuit shown in the figure. Find charge on each capacitor

1) $q_1 = q_3 = 0; q_2 = q_4 = CV$
2) $q_1 = q_3 = 0; q_2 = q_4 = 0$
3) $q_1 = q_3 = CV; q_2 = q_4 = CV$
4) None of these

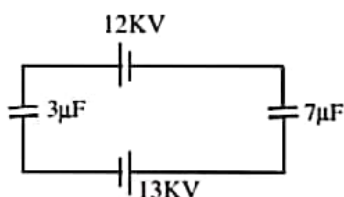


30. Find the potential difference between the points A and B if $E = 110\text{V}$ and the capacitance ratio $C_2/C_1 = 2$



1) 10 V 2) 9 V 3) 5 V 4) 4 V

31. In the circuit diagram given below, the value of the potential difference across the plates of the capacitors are

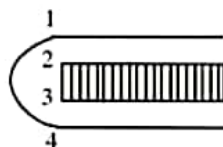


- 1) 75 KV 2) 70 KV 3) 7 KV 4) 7.5 KV

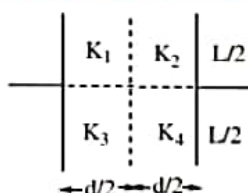
Di electrics :

32. A parallel plate capacitor has capacitance of $2\mu\text{F}$ in air and $12\mu\text{F}$ when immersed in oil. The dielectric constant of the oil is
 1) 5 2) 4 3) 3 4) 6
33. When a dielectric slab of thickness 6 cm is introduced between the plates of parallel plate condenser, it is found that the distance between the plates has to be increased by 4 cm to restore to capacity to original value. The dielectric constant of the slab is
 1) 1.5 2) $\frac{2}{3}$ 3) 3 4) 4
34. Two identical parallel plate capacitors are joined in series to 100 V battery. Now a dielectric with $K = 4$ is introduced between the plates of second capacitor. The potential difference on capacitors are
 1) 60 V, 40 V 2) 70 V, 30 V 3) 75 V, 25 V 4) 80 V, 20 V
35. A capacitor connected to a 10V battery collects a charge $40\mu\text{C}$ with air as dielectric and $100\mu\text{C}$ with oil as dielectric. The dielectric constant of oil is
 1) 2 2) 2.5 3) 4 4) 10
36. A parallel plate capacitor with air as medium between the plates has a capacitance 10 mF. The area of the capacitor is divided into two equal halves and filled with two media having dielectric constants $K_1 = 2$ and $K_2 = 4$. The capacitance of the system will be
 1) 10 mF 2) 20 mF 3) 30 mF 4) 40 mF
37. A parallel plate capacitor has the space between its plates filled by two slabs of thickness $\frac{d}{2}$ each and dielectric constant K_1 and K_2 . d is the plate separation of the capacitor. The capacitance of the capacitor is
 1) $C_s = \frac{k_1 k_2}{k_1 + k_2} C$ 2) $C_s = \frac{2k_1 k_2}{k_1 + k_2} C$ 3) $C_s = \frac{k_1 + k_2}{k_1 k_2} C$ 4) none
38. A parallel plate capacitor with air between the plates has a capacitance of 9 pF. The separation between its plates is 'd'. The space between the plates is now filled with two dielectrics. One of the dielectrics has dielectric constant $k_1 = 3$ and thickness $\frac{d}{3}$ while the other one has dielectric constant $k_2 = 6$ and thickness $\frac{2d}{3}$. Capacitance of the capacitor is now
 1) 1.8pF 2) 45 pF 3) 40.5 pF 4) 20.25 pF

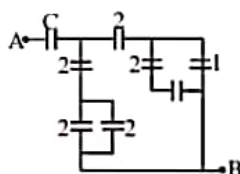
39. Four identical plates 1, 2, 3 and 4 are placed parallel to each other at equal distance as shown in the figure. Plates 1 and 4 are joined together and the space between 2 and 3 is filled with a dielectric of dielectric constant $k = 2$. The capacitance of the system between 1 and 3 & 2 and 4 are C_1 and C_2 respectively. The ratio $\frac{C_1}{C_2}$ is



- 1) $\frac{5}{3}$ 2) 1 3) $\frac{3}{5}$ 4) $\frac{5}{7}$
40. A parallel plate capacitor with square plates is filled with four dielectrics of dielectric constants K_1, K_2, K_3, K_4 arranged as shown in the figure. The effective dielectric constant K will be



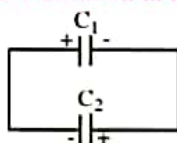
- 1) $K = \frac{(K_1 + K_2)(K_3 + K_4)}{2(K_1 + K_2 + K_3 + K_4)}$ 2) $K = \frac{(K_1 + K_2)(K_3 + K_4)}{(K_1 + K_2 + K_3 + K_4)}$
- 3) $K = \frac{(K_1 + K_4)(K_2 + K_3)}{2(K_1 + K_2 + K_3 + K_4)}$ 4) $K = \frac{(K_1 + K_3)(K_2 + K_4)}{(K_1 + K_2 + K_3 + K_4)}$
41. Voltage rating of a parallel plate capacitor is 500 V. Its dielectric can withstand a maximum electric field of 10^6 V/m. The plate area is 10^{-4} m². What is the dielectric constant if the capacitance is 15 pF? (given $\epsilon_0 = 8.86 \times 10^{-12}$ C² / Nm²)
- 1) 6.2 2) 4.5 3) 3.8 4) 8.5
42. In the circuit shown, find C if the effective capacitance of the whole circuit is to be $0.5 \mu\text{F}$. All values in the circuit are in μF .



- 1) $\frac{7}{11} \mu\text{F}$ 2) $\frac{6}{5} \mu\text{F}$ 3) $4 \mu\text{F}$ 4) $\frac{7}{10} \mu\text{F}$
43. The distance between the plates of a condenser is reduced to $\frac{1}{4}$ th and the space between the plates is filled up by a medium of dielectric constant $K = 2.8$. The capacity is increased to
- 1) 11.2 2) 22 times 3) 14 times 4) none

Regrouping & Heat produced problems :

44. A condenser of capacity 16 mF charged to a potential of 20 V is connected to a condenser of capacity C charged to a potential of 10 V as shown in the figure.



If the common potential is 14 V, the capacity C is equal to

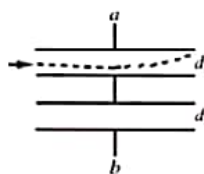
- 1) 18 mF 2) 24 mF 3) 8 mF 4) 4 mF
45. Given a number of capacitors labelled as $8\mu\text{F}$, 250V. Find the minimum number of capacitors needed to get an arrangement equivalent to $16\mu\text{F}$, 1000V
- 1) 4 2) 16 3) 32 4) 64
46. A parallel plate condenser has plates of area 200 cm^2 and separation 0.05 cm. The space between plates have been filled with a dielectric having $K = 8$ and then charged to 300 volts. The stored energy
- 1) $1.6 \times 10^{-5}\text{ J}$ 2) $2 \times 10^{-6}\text{ J}$ 3) $12.4 \times 10^{-5}\text{ J}$ 4) $1.6 \times 64 \times 10^{-5}\text{ J}$
47. Two spheres of radii 12 cm and 16 cm have equal charge. The ratio of their energies is
- 1) 3 : 4 2) 4 : 3 3) 1 : 2 4) 2 : 1
48. The potential difference between the plates of a condenser is increased by 20%. The energy stored in the condenser increases by
- 1) 50% 2) 400% 3) 44% 4) 30% nearly
49. The equivalent capacity of four identical capacitors is $12\mu\text{F}$ when they are connected in parallel and a potential of 400 V is applied. If they are connected in series and same potential is applied then change in energy stored in the system is
- 1) $90 \times 10^{-2}\text{ J}$ 2) $48 \times 10^{-2}\text{ J}$ 3) $40 \times 10^{-3}\text{ J}$ 4) $20 \times 10^{-2}\text{ J}$
50. A $2\mu\text{F}$ condenser is charged to 500V and then the plates are joined through a resistance. The heat produced in the resistance in joule is
- 1) $2.5 \times 10^{-2}\text{ J}$ 2) $20 \times 10^{-2}\text{ J}$ 3) $12 \times 10^{-2}\text{ J}$ 4) $25 \times 10^{-2}\text{ J}$
51. A 20F capacitor is charged to 5V and isolated. It is then connected in parallel with an uncharged 30F capacitor. The decrease in the energy of the system will be (2001 E-1)
- 1) 25 J 2) 100 J 3) 125 J 4) 150 J
52. A $0.2\mu\text{F}$ capacitor is charged to 600 V. After disconnecting the battery, it is put in parallel with an uncharged $1\mu\text{F}$ capacitor. The potential now is
- 1) 100 V 2) 200 V 3) 150 V 4) 10 V
53. A $100\mu\text{F}$ capacitor is charged to 200 volt. It is discharged through a 2 ohm resistance. The amount of heat generated will be
- 1) 2 J 2) 4 J 3) 0.2 J 4) 0.4 J

Miscellaneous :

54. Both the capacitors shown in figure are made of square plates of edge a . The separation between the plates of the capacitors are d_1 and d_2 as shown in the figure. A potential difference V is applied between the points a and b . An electron is projected between the plates of the upper capacitor along the central line. With what minimum speed should the electron be projected so that it does not collide with any plate ? Consider only the electric forces.

$$1) \left[\frac{Vea^2}{md_2(d_1 - d_2)} \right]^{1/2} \quad 2) \left[\frac{Vea^2}{md_1(d_1 + d_2)} \right]^{1/2}$$

$$3) \left[\frac{Vea^2}{md_2(d_1 - d_2)} \right]^2 \quad 4) \left[\frac{Vea^2}{md_1(d_1 + d_2)} \right]^2$$



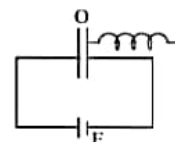
55. If an electron enters into a space between the plates of a parallel plate capacitor at an angle α with the plates and leaves at angle β to the plates. The ratio of its kinetic energy while entering the capacitor to that while leaving will be :

$$1) \left[\frac{\cos \alpha}{\cos \beta} \right]^2 \quad 2) \left[\frac{\cos \beta}{\cos \alpha} \right]^2 \quad 3) \left[\frac{\sin \alpha}{\sin \beta} \right]^2 \quad 4) \left[\frac{\sin \beta}{\sin \alpha} \right]^2$$

56. One plate of a capacitor is connected to a spring as shown. Area of both the plate is A and separation is d when capacitor is uncharged. When capacitor is charged, in steady state separation between the plates is $0.8d$. The force constant of the spring is approximately :

$$1) \frac{\epsilon_0 AE^2}{2d^3} \quad 2) \frac{2\epsilon_0 AE}{d^2}$$

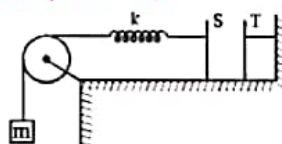
$$3) \frac{6\epsilon_0 E^2}{Ad^3} \quad 4) \frac{4\epsilon_0 AE^2}{d^3}$$



57. The plates S and T of an uncharged parallel plate capacitor are connected across a battery. The battery is then disconnected and the charged plates are now connected in a system as shown in the figure. The system shown is in equilibrium. All the strings are insulating and massless. The magnitude of charge on one of the capacitor plates is: [Area of plates = A]

$$1) \sqrt{2mgA\epsilon_0} \quad 2) \sqrt{\frac{4mgA\epsilon_0}{k}}$$

$$3) \sqrt{mgA\epsilon_0} \quad 4) \sqrt{\frac{2mgA\epsilon_0}{k}}$$



58. The capacitance of a parallel plate condenser, with plate area A and distance between plates d , when filled with a dielectric whose dielectric constant varies as, is $\epsilon(x) = \epsilon_0 + \beta x$, for $0 < x < \frac{d}{2}$
 $\epsilon(x) = \epsilon_0 + \beta(d - x)$ for $\frac{d}{2} < x < d$

$$1) \frac{\beta \frac{A}{2}}{\log_e \left(\frac{\epsilon_0}{\beta d} \right)} \quad 2) \frac{A}{2} \log_e \left(\frac{\epsilon_0}{\beta d} \right)$$

$$3) \frac{A}{2} \log_e \left(\frac{\epsilon_0}{\epsilon_0 - \frac{\beta d}{2}} \right) \quad 4) \frac{\beta \frac{A}{2} \epsilon_0}{\log_e \left(\frac{\epsilon_0 + \frac{\beta d}{2}}{\epsilon_0} \right)}$$

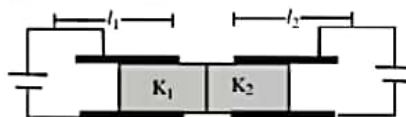
59. Figure shows two parallel plate capacitors with fixed plates and connected to two batteries. The separation between the plates is the same for the two capacitors. The plates are rectangular in shape with width b and lengths l_1 and l_2 . The left half of the dielectric slab has a dielectric constant K_1 and the right half K_2 . Neglecting any friction, find the ratio of the emf of the left battery to that of the right battery for which the dielectric slab may remain in equilibrium

1) $\sqrt{\frac{K_2-1}{K_1-1}}$

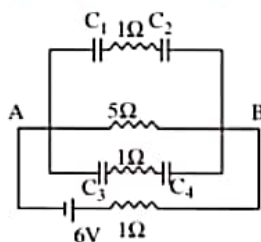
2) $\sqrt{\frac{K_1-1}{K_2-1}}$

3) $\sqrt{\frac{K_1+1}{K_2+1}}$

4) $\sqrt{\frac{K_2+1}{K_1+1}}$



60. In the figure shown, each capacitor is $1\mu\text{F}$ and the resistors are indicated. The charges Q_1 and Q_4 on the capacitors C_1 and C_4 are respectively under steady condition. Then,



1) $Q_1 > Q_4$

2) $Q_1 = 2.5\mu\text{C}$

3) $Q_1 < Q_4$

4) $Q_4 = 5\mu\text{C}$

Previous Main Problems :

61. A capacitance of $2\mu\text{F}$ is required in an electrical circuit across a potential difference of 1.0kV . A large number of $1\mu\text{F}$ capacitors are available which can withstand a potential difference of not more than 300 V . The minimum number of capacitors required to achieve this is [2017]

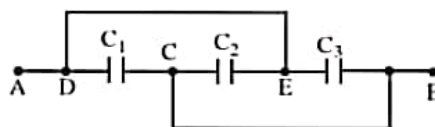
1) 24

2) 32

3) 2

4) 16

62. A combination of parallel plate capacitors is maintained at a certain potential difference.



When a 3 mm thick slab is introduced between all the plates, in order to maintain the same potential difference, the distance between the plates is increased by 2.4 mm .

Find the dielectric constant of the slab.

[Online April 9, 2017]

1) 3

2) 4

3) 5

4) 6

63. The energy stored in the electric field produced by a metal sphere is 4.5J . If the sphere contains $4\mu\text{C}$ charge, its radius will be: [Take : $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9\text{ N-m}^2/\text{C}^2$] [Online April 8, 2017]

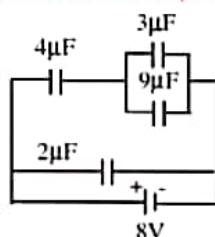
1) 20mm

2) 32mm

3) 28mm

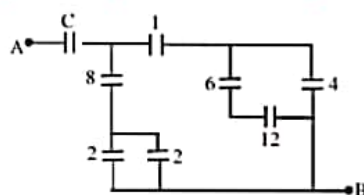
4) 16 mm

64. A combination of capacitors is set up as shown in the figure. The magnitude of the electric field, due to a point charge Q (having a charge equal to the sum of the charges on the $4\mu\text{F}$ and $9\mu\text{F}$ capacitors), at a point distance 30m from it, would equal [2016]



- 1) 420 N/C 2) 480 N/C 3) 240 N/C 4) 360 N/C

65. Figure shows a network of capacitors where the numbers indicate capacitances in micro Farad. The value of capacitance C if the equivalent capacitance between point A and B is to be $1\mu\text{F}$ is : [Online April 10, 2016]

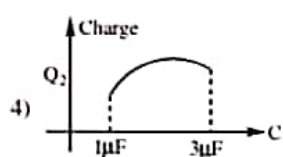
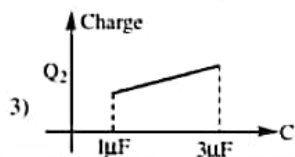
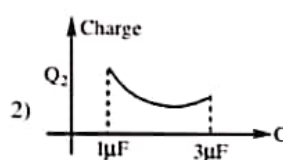
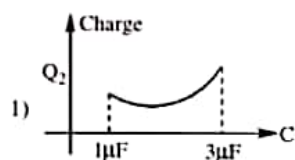
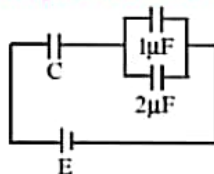


- 1) $\frac{32}{23}\mu\text{F}$ 2) $\frac{31}{23}\mu\text{F}$ 3) $\frac{33}{23}\mu\text{F}$ 4) $\frac{34}{23}\mu\text{F}$

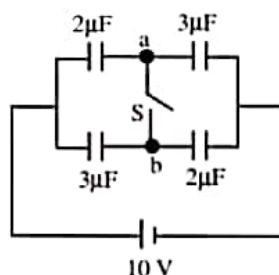
66. Three capacitors each of $4\mu\text{F}$ are to be connected in such a way that the effective capacitance is $6\mu\text{F}$. This can be done by connecting them: [Online April 9, 2016]

- 1) all in series 2) all in parallel
3) two in parallel and one in series 4) two in series and one in parallel

67. In the given circuit, charge Q_2 on the $2\mu\text{F}$ capacitor changes as C is varied from $1\mu\text{F}$ to $3\mu\text{F}$. Q_2 as a function of ' C ' is given properly by: (figures are drawn schematically and are not to scale) [2015]



68. In figure a system of four capacitors connected across a 10 V battery is shown. Charge that will flow from switch S when it is closed is [Online April 11, 2015]



- 1) $5\mu\text{C}$ from b to a 2) $20\mu\text{C}$ from a to b 3) zero 4) $5\mu\text{C}$ from a to b
69. A parallel plate capacitor is made of two circular plates separated by a distance 5 mm and with a dielectric of dielectric constant 2.2 between them. When the electric field in the dielectric is $3 \times 10^4 \text{ V/m}$ the charge density of the positive plate will be close to : [2014]
- 1) $6 \times 10^{-7} \text{ C/m}^2$ 2) $3 \times 10^{-7} \text{ C/m}^2$ 3) $3 \times 10^4 \text{ C/m}^2$ 4) $6 \times 10^4 \text{ C/m}^2$
70. The gap between the plates of a parallel plate capacitor of area A and distance between plates d, is filled with a dielectric whose permittivity varies linearly from ϵ_1 at one plate to ϵ_2 at the other. The capacitance of capacitor is : [Online April 19, 2014]
- 1) $\epsilon_0 (\epsilon_1 + \epsilon_2) A / d$ 2) $\epsilon_0 (\epsilon_2 + \epsilon_1) A / 2d$
 3) $\epsilon_0 A / [d \ln(\epsilon_2 / \epsilon_1)]$ 4) $\epsilon_0 (\epsilon_1 - \epsilon_2) A / [d \ln(\epsilon_2 / \epsilon_1)]$
71. The space between the plates of a parallel plate capacitor is filled with a 'dielectric' whose 'dielectric constant' varies with distance as per the relation : $K(x) = K_0 + \lambda x$ ($\lambda = \text{a constant}$)
 The capacitance C, of the capacitor, would be related to its vacuum capacitance C_0 for the relation : [Online April 12, 2014]

1) $C = \frac{\lambda d}{\ln(1 + K_0 \lambda d)} C_0$

2) $C = \frac{\lambda}{d \ln(1 + K_0 \lambda d)} C_0$

3) $C = \frac{\lambda d}{\ln(1 + \lambda d / K_0)} C_0$

4) $C = \frac{\lambda}{d \ln(1 + K_0 / \lambda d)} C_0$

72. A parallel plate capacitor is made of two plates of length l, width w and separated by distance d. A dielectric slab (dielectric constant K) that fits exactly between the plates is held near the edge of the plates. It is pulled into the capacitor by a force $F = -\frac{\partial U}{\partial x}$ where U is the energy of the capacitor when dielectric is inside the capacitor up to distance x (See figure). If the charge on the capacitor is Q then the force on the dielectric when it is near the edge is : [Online April 11, 2014]

1) $\frac{Q^2 d}{2\omega l^2 \epsilon_0} K$

2) $\frac{Q^2 \omega}{2dl^2 \epsilon_0} (K-1)$

3) $\frac{Q^2 d}{2\omega l^2 \epsilon_0} (K-1)$

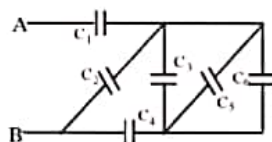
4) $\frac{Q^2 \omega}{2dl^2 \epsilon_0} K$



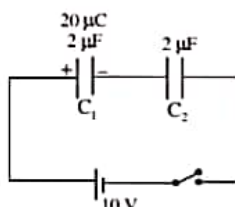
73. Three capacitors, each of $3\mu\text{F}$, are provided. These cannot be combined to provide. These cannot be combined to provide the resultant capacitance of : [Online April 9, 2014]
 1) $1\mu\text{F}$ 2) $2\mu\text{F}$ 3) $4.5\mu\text{F}$ 4) $6\mu\text{F}$
74. A parallel plate capacitor having a separation between the plates d , plate area A and material with dielectric constant K has capacitance C_0 . Now one-third of the material is replaced by another material with dielectric constant $2K$, so that effectively there are two capacitors one with area $\frac{1}{3}A$, dielectric constant $2K$ and another with area $\frac{2}{3}A$ and dielectric constant K . If the capacitance of this new capacitor is C then $\frac{C}{C_0}$ is [Online April 25, 2013]
 1) 1 2) $\frac{4}{3}$ 3) $\frac{2}{3}$ 4) $\frac{1}{3}$
75. To establish an instantaneous current of 2A through a $1\mu\text{F}$ capacitor ; the potential difference across the capacitor plates should be changed at the rate of [Online April 22, 2013]
 1) $2 \times 10^4 \text{ V/s}$ 2) $4 \times 10^6 \text{ V/s}$ 3) $2 \times 10^6 \text{ V/s}$ 4) $4 \times 10^4 \text{ V/s}$

Numerical Value Type Questions

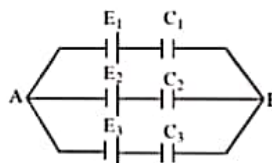
76. 12 identical capacitors are connected in series between two points. Out of these for 'n' capacitors, the spacing between the plates is reduced to half and for the remaining it is doubled. What is the value of 'n' for which the effective capacitance remains unchanged?
77. In the accompanying diagram, if $C_1 = 3\mu\text{F}$, $C_2 = 6\mu\text{F}$, $C_3 = 9\mu\text{F}$, $C_4 = 12\mu\text{F}$, $C_5 = 15\mu\text{F}$ and $C_6 = 18\mu\text{F}$, then the equivalent capacitance between the ends A and B is (in μF)



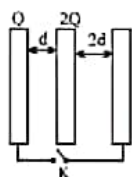
78. When switch is closed, charge on C_1 is _____ μC



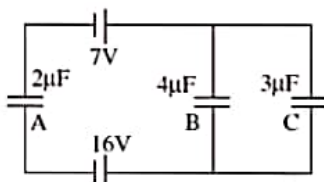
79. In the given circuit find the $|V_A - V_B|$ where V_A, V_B are electric potentials at the points A and B. ($E_1 = 1\text{V}$, $E_2 = 2\text{V}$, $E_3 = 3\text{V}$, $C_1 = 1\mu\text{F}$, $C_2 = 2\mu\text{F}$, $C_3 = 3\mu\text{F}$)



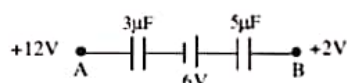
80. Three large plates are arranged as shown. How much charge will flow through the key k if it is closed? (if $Q = 1C$)



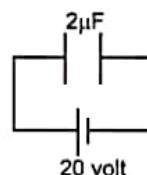
81. A parallel plate capacitor is charged by a supply of 500V. Plate separation is 1mm. If capacitor is lowered into water with water filling in the gap between the plates, then change in pressure at any point between the plates is $K\epsilon_0 \times 10^{13} (N/m)^2$. Find the value of K , given ϵ_r (for water)=81.
82. Three uncharged capacitors A, B and C are connected as shown in the figure. The charge on capacitor B is , in the steady state, (in μC)



83. A and B are two points in a closed circuit. The potential difference across the condenser of capacity $5\mu F$ is



84. In the figure a capacitor of capacitance $2\mu F$ is connected to a cell of emf 20 volt. The plates of the capacitor are drawn apart slowly to double the distance between them. The work done by the external agent on the plates is : (of the order of $10^{-2} J$)



85. Twelve equal capacitors are joined to form a cube. The equivalent capacitance between two opposite edges on the same plane is _____ times of 'C'.

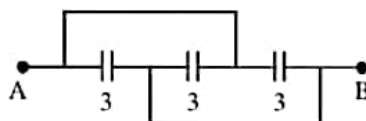
EXERCISE - II

Straight Objective Type Questions

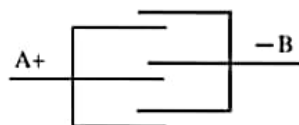
Capacitors and its combination :

- The charge stored in a capacitor is $20\mu C$ and the potential difference across the plates is 500V. Its capacity is
 1) $0.4\mu F$ 2) $0.04\mu F$ 3) $4\mu F$ 4) $40\mu F$
- A sphere A of radius 5 cm contains a charge of 120 nC. It is connected to two other spheres B and C having respective radii of 10 cm and 15 cm. Determine the charges on each of the spheres.
 1) 20nC, 60nC, 40nC 2) 20nC, 40nC, 60nC 3) 20nC, 40nC, 10nC 4) 20nC, 80nC, 20nC

3. The capacitance of a sphere of radius 10cm situated in air is approximately
 1) 1×10^{-2} F 2) 11×10^{-12} F 3) 111×10^{-12} F 4) 15×10^{-12} F
4. Sixty four spherical drops each of radius 2 cm and carrying 5C charge combine to form a bigger drop. Its capacity is
 1) $\frac{9}{8} \times 10^{-11}$ F 2) $\frac{1}{8} \times 10^{-11}$ F 3) $\frac{8}{9} \times 10^{-11}$ F 4) $\frac{1}{9} \times 10^{-11}$ F
5. The capacity of a parallel plate condenser consisting of two plates each 10 cm square and are separated by a distance of 2 mm is (Take air as the medium between the plates)
 1) 44.25×10^{-12} F 2) 4.425×10^{-12} F 3) 44.25×10^{-12} F 4) 4.425×10^{-12} F
6. The ratio of the resultant capacities when three capacitors of $2\mu\text{F}$, $4\mu\text{F}$ and $6\mu\text{F}$ are connected first in series and then in parallel is
 1) 1:2 2) 2:1 3) 11:1 4) 1:11
7. A conductor A of capacity $4\mu\text{F}$ has a charge $20\mu\text{F}$ and another condenser B of capacity $10\mu\text{F}$ has a charge $40\mu\text{F}$. If they are connected parallel, then
 1) charge flows from B to A till the charges on them are equal
 2) charge flows from B to A till common potential is reached
 3) charge flows from A to B till common potential is reached
 4) charge flows from A to B till charges on them are equal
8. When two capacitors are joined in the resultant capacity is $2.4\mu\text{F}$ and when the same two are joined in parallel the resultant capacity is $10\mu\text{F}$. Their individual capacities are
 1) $C_1 = 5\mu\text{F}$, $C_2 = 2\mu\text{F}$ 2) $C_1 = 5\mu\text{F}$, $C_2 = 5\mu\text{F}$
 3) $C_1 = 6\mu\text{F}$, $C_2 = 1\mu\text{F}$ 4) $C_1 = 6\mu\text{F}$, $C_2 = 4\mu\text{F}$
9. The equivalent capacity (in μF) between the points 'A' and 'B' in the following figure will be

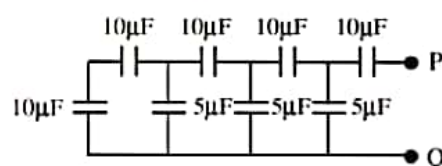


- 1) $18\mu\text{F}$ 2) $10\mu\text{F}$ 3) $16\mu\text{F}$ 4) $2\mu\text{F}$
10. A radio capacitor of variable capacitance is made of n parallel plates each of area A and separated from each other by a distance d . The alternate plates are connected together. The capacitance of the combination is



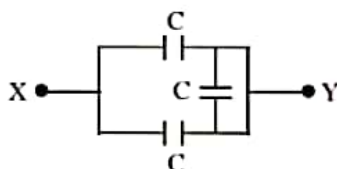
- 1) $\frac{(n-1)\epsilon_0 A}{d}$ 2) $\frac{n\epsilon_0 A}{d}$ 3) $\frac{\epsilon_0 A}{d}$ 4) none

11. The equivalent capacitance between P and Q is



- 1) $10\ \mu\text{F}$ 2) $12\ \mu\text{F}$ 3) $5\ \mu\text{F}$ 4) $15\ \mu\text{F}$

12. The equivalent capacity between the points X and Y in the circuit with $C = 1\ \mu\text{F}$

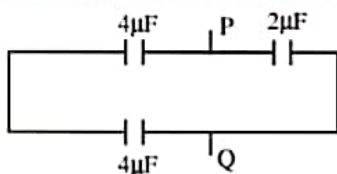


- 1) $2\ \mu\text{F}$ 2) $4\ \mu\text{F}$ 3) $1\ \mu\text{F}$ 4) $10\ \mu\text{F}$

13. Two capacitors with capacitances C_1 and C_2 are charged to potentials V_1 and V_2 respectively. When they are connected in parallel the ratio of their respective charges is

- 1) $\frac{C_1}{C_2} = \frac{V_1}{V_2}$ 2) $\frac{C_2}{C_1} = \frac{V_1}{V_2}$ 3) $\frac{Q_1}{Q_2} = \frac{C_1}{C_2}$ 4) $\frac{Q_2}{Q_1} = \frac{C_1}{C_2}$

14. The resultant capacity between the points P and Q of the given figure is

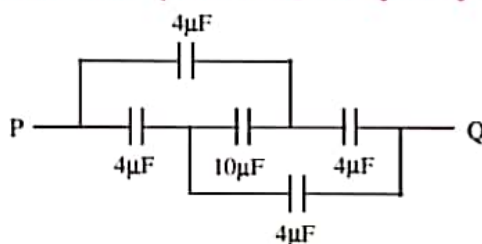


- 1) $2\ \mu\text{F}$ 2) $4\ \mu\text{F}$ 3) $6\ \mu\text{F}$ 4) $8\ \mu\text{F}$

15. Three capacitors of capacitance $3\ \mu\text{F}$ are connected in a circuit. Then their maximum and minimum capacitances will be

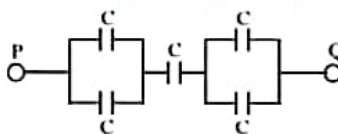
- 1) $9\ \mu\text{F}, 1\ \mu\text{F}$ 2) $8\ \mu\text{F}, 2\ \mu\text{F}$ 3) $9\ \mu\text{F}, 0\ \mu\text{F}$ 4) $3\ \mu\text{F}, 2\ \mu\text{F}$

16. The effective capacitance between the point P and Q in the given figure is



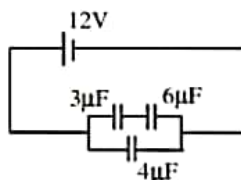
- 1) $4\ \mu\text{F}$ 2) $14\ \mu\text{F}$ 3) $2\ \mu\text{F}$ 4) $12\ \mu\text{F}$

17. The equivalent capacitance between P and Q of the given figure is (the capacitance of each capacitor is $1\mu\text{F}$)



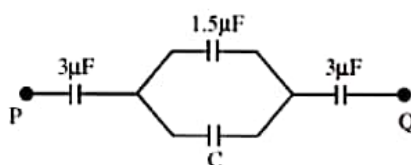
- 1) $5\mu\text{F}$ 2) $50\mu\text{F}$ 3) $15\mu\text{F}$ 4) $0.5\mu\text{F}$

18. Charge 'Q' taken from the battery of 12 V in the circuit is



- 1) $72\mu\text{C}$ 2) $17\mu\text{C}$ 3) $82\mu\text{C}$ 4) $80\mu\text{C}$

19. The equivalent capacitance of the network given below is $1\mu\text{F}$. The value of 'C' is



- 1) $10\mu\text{F}$ 2) $1.5\mu\text{F}$ 3) $2.5\mu\text{F}$ 4) $0.5\mu\text{F}$

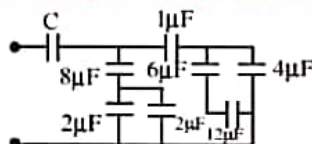
20. Three capacitors of capacitance $2\mu\text{F}$, $4\mu\text{F}$ and $6\mu\text{F}$ are connected in parallel and a potential difference of 12 V is applied. The charge on each capacitor

- 1) $6\mu\text{C}$, $2\mu\text{C}$, $1\mu\text{C}$ 2) $3\mu\text{C}$, $6\mu\text{C}$, $18\mu\text{C}$
3) $24\mu\text{C}$, $48\mu\text{C}$, $72\mu\text{C}$ 4) $7\mu\text{C}$, $14\mu\text{C}$, $21\mu\text{C}$

21. Three capacitors of $3\mu\text{F}$, $2\mu\text{F}$ and $6\mu\text{F}$ are connected in series. When a battery of 10V is connected to this combination then charge on $3\mu\text{F}$ capacitor will be

- 1) $1\mu\text{F}$ 2) $5\mu\text{F}$ 3) $10\mu\text{F}$ 4) $15\mu\text{F}$

22. In the following circuit, the resultant capacitance between A and B is $1\mu\text{F}$. Find the value of C.

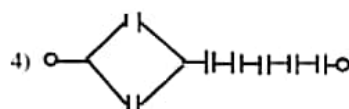
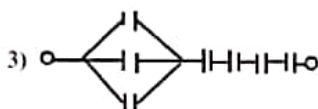
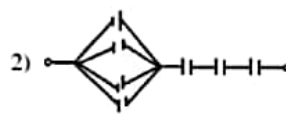


- 1) $\frac{32}{23}\mu\text{F}$ 2) $\frac{11}{23}\mu\text{F}$ 3) $6\mu\text{F}$ 4) $1\mu\text{F}$

23. Two condensers of capacity C and $2C$ are connected in parallel and these are charged upto V volt. If the battery is removed and dielectric medium of constant K is put between the plates of first condenser, then the potential at each condenser is

- 1) $\frac{V}{K+1} = V^1$ 2) $\frac{K+1}{2} = V^1$ 3) $\frac{3V}{K+2} = V^1$ 4) $\frac{V}{3K} = V^1$

24. Seven capacitors each of capacitance $2\mu\text{F}$ are to be connected in a configuration to obtain an effective capacitance of $\frac{10}{11}\mu\text{F}$. Which of the combination shown in figure will achieve the desired result



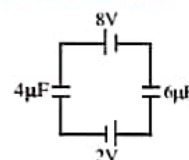
25. Two capacitors of capacities $3\mu\text{F}$ and $6\mu\text{F}$ are connected series and connected to 120V . The potential differences across $3\mu\text{F}$ is V_0 and the charge here is q_0 . We have

- A) $q_0 = 40\mu\text{C}$ B) $V_0 = 60\text{V}$ C) $V_0 = 80\text{V}$ D) $q_0 = 240\mu\text{C}$
 1) 8V 2) 18V 3) 10V 4) 80V

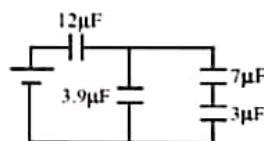
Capacitor circuits :

26. Two condensers of capacities $4\mu\text{F}$ and $6\mu\text{F}$ are connected to two cells as shown. Then

- 1) P.D. of $6\mu\text{F}$ condenser is 4V
 2) charge of $4\mu\text{F}$ condenser is $24\mu\text{C}$
 3) both of the above are true
 4) none of the above is true

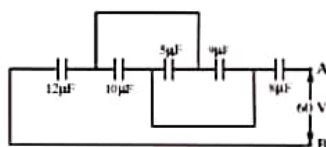


27. The potential drop across $7\mu\text{F}$ capacitor is 6V . Then



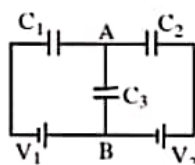
- 1) potential drop across $3\mu\text{F}$ capacitor is 10V 2) charge on $3\mu\text{F}$ capacitor is $21\mu\text{C}$
 3) emf of the cell is 30V 4) P.D across $12\mu\text{F}$ capacitor is 5V

28. The energy stored in $5\mu\text{F}$ and $8\mu\text{F}$ capacitors are



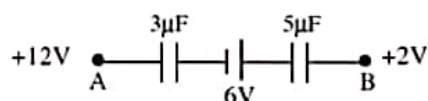
- 1) $250 \times 10^{-6}\text{ J}$, $36 \times 10^{-4}\text{ J}$ 2) $250 \times 10^{-6}\text{ J}$, $240 \times 10^{-4}\text{ J}$
 3) $250 \times 10^{-6}\text{ J}$, $240 \times 10^{-4}\text{ J}$ 4) $250 \times 10^{-6}\text{ J}$, $50 \times 10^{-4}\text{ J}$

29. Find the potential difference $V_A - V_B$



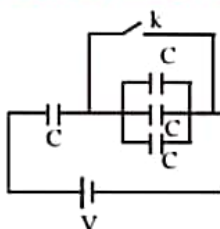
- 1) $\frac{V_2 C_2 - V_1 C_1}{C_1 + C_2 + C_3}$ 2) $\frac{V_2 C_2 + V_1 C_1}{C_1 + C_2 + C_3}$ 3) $\frac{V_1 C_2 + V_2 C_1}{C_1 + C_2 + C_3}$ 4) $\frac{V_1 C_2 - V_2 C_1}{C_1 + C_2 + C_3}$

30. A and B are two points in a closed circuit. The potential difference across the condenser of capacity $5\mu\text{F}$ is



- 1) 6V 2) 10V 3) 16V 4) 4V

31. The charge flowing through the cell on closing the key k is equal to



- 1) $\frac{CV}{2}$ 2) $\frac{CV}{4}$ 3) $\frac{CV}{5}$ 4) CV

Di electrics :

32. An oil condenser has a capacity of $100\mu\text{F}$. The oil has dielectric constant 2. When the oil leaks out its new capacity is

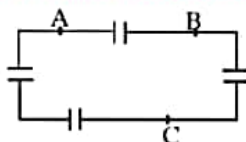
- 1) $10\mu\text{F}$ 2) $50\mu\text{F}$ 3) $20\mu\text{F}$ 4) $100\mu\text{F}$

33. When a dielectric slab of thickness 4 cm is introduced between the plates of parallel plate condenser, it is found the distance between the plates has to be increased by 3cm to restore to capacity to original value. The dielectric constant of the slab is

- 1) $K = 40$ 2) $K = 25$ 3) $K = 2$ 4) $K = 4$

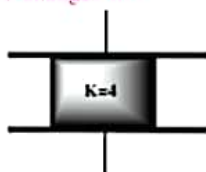
34. Four capacitors of capacity $3\mu\text{F}$ are connected as shown in figure. The ratio of equivalent capacitance between A and B and between A and C will be

- 1) 4 : 3
2) 3 : 4
3) 2 : 3
4) 3 : 2



35. Consider a parallel plate capacitor of $10\mu\text{F}$ with air filled in the gap between the plates. Now one half of the space between the plates is filled with a dielectric of dielectric constant 4, as shown in the figure. The capacity of the capacitor changes to

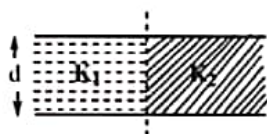
- 1) $25\mu\text{F}$
 2) $20\mu\text{F}$
 3) $40\mu\text{F}$
 4) $5\mu\text{F}$



36. The area of the positive plate is A_1 and the area of the negative plate is A_2 ($A_2 < A_1$). They are parallel to each other and are separated by a distance d . The capacity of a condenser with air as dielectric is

- 1) $C' = \frac{C}{d}$ 2) $C' = \frac{A_2 \epsilon_0}{d}$ 3) $C' = \frac{\epsilon_0 A}{d}$ 4) $C' = \frac{d}{A \epsilon_0}$

37. A parallel plate capacitor with plate area 'A' and separation 'd' is filled with two dielectrics of dielectric constants K_1 and K_2 . If the permittivity of free space is ϵ_0 , the capacitance of the capacitor is given by

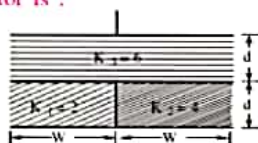


- 1) $\frac{A \epsilon_0}{2d} (K_1 + K_2)$ 2) $\frac{A \epsilon_0}{2d} (2K)$ 3) $\frac{\epsilon_0}{2d} (K_1 + K_2)$ 4) $\frac{A \epsilon_0}{d} (K_1 + K_2)$

38. A capacitor connected to a 10V battery collects a charge $40\mu\text{C}$ with air as dielectric and $100\mu\text{C}$ with oil as dielectric. The dielectric constant of oil is

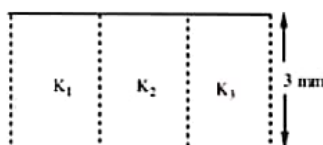
- 1) 2 2) 2.5 3) 4 4) 10

39. A parallel plate capacitor C (without dielectric) is filled by dielectric slabs as shown in figure. Then the new capacitance of the capacitor is :



- 1) 3.9 C 2) 4 C 3) 2.4 C 4) 3 C

40. A parallel plate capacitor is of area 6cm^2 and a separation 3 mm. The gap is filled with three dielectric materials of equal thickness (see figure) with dielectric constants $K_1=10$, $K_2=12$ and $K_3=14$. The dielectric constant of a material which when fully inserted in above capacitor, gives same capacitance would be



- 1) 4 2) 14 3) 12 4) 36

41. Two identical parallel plate capacitors, of capacitance C each, have plates of area A , separated by a distance d . The space between the plates of the two capacitors, is filled with three dielectrics, of equal thickness and dielectric constants K_1 , K_2 and K_3 . The first capacitor is filled as shown in fig. I, and the second one is filled as shown in fig II. If these two modified capacitors are charged by the same potential V , the ratio of the energy stored in the two, would be (E_1 refers to capacitor (I) and E_2 to capacitor (II)).

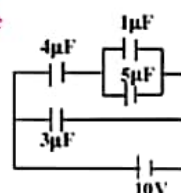


$$1) \frac{E_1}{E_2} = \frac{(K_1 + K_2 + K_3)(K_2 K_3 + K_3 K_1 + K_1 K_2)}{K_1 K_2 K_3} \quad 2) \frac{E_1}{E_2} = \frac{K_1 K_2 K_3}{(K_1 + K_2 + K_3)(K_2 K_3 + K_3 K_1 + K_1 K_2)}$$

$$3) \frac{E_1}{E_2} = \frac{(K_1 + K_2 + K_3)(K_2 K_3 + K_3 K_1 + K_1 K_2)}{9 K_1 K_2 K_3} \quad 4) \frac{E_1}{E_2} = \frac{9 K_1 K_2 K_3}{(K_1 + K_2 + K_3)(K_2 K_3 + K_3 K_1 + K_1 K_2)}$$

42. In the given circuit, the charge on $4\mu\text{F}$ capacitor will be

- 1) $9.6\mu\text{C}$
2) $24\mu\text{C}$
3) $5.4\mu\text{C}$
4) $13.4\mu\text{C}$

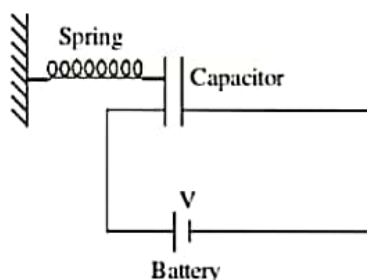


43. A parallel plate capacitor has a capacity $2\mu\text{F}$. A dielectric slab ($K = 5$) is inserted between the plates and capacitor is charged with 100 V and then isolated. The potential when dielectric is removed is
- 1) 200 V 2) 500 V 3) 300 V 4) 10 V

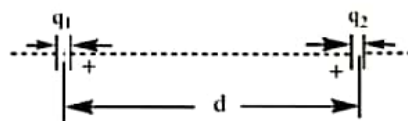
Regrouping & Heat produced problems :

44. A capacitor of $30\mu\text{F}$ charged to 100 V is connected in parallel to capacitor of $20\mu\text{F}$ charged to 50 volt . The common potential is
- 1) 0.8V 2) 18V 3) 8V 4) 80V
45. 'A' and 'B' are two condensers of capacities $2\mu\text{F}$ and $4\mu\text{F}$. They are charged to potential differences of 12V and 6V respectively. If they are now connected (+ve to +ve), the charge that flows through the connecting wire is
- 1) $8\mu\text{C}$ from A to B 2) $8\mu\text{C}$ from B to A 3) $2\mu\text{C}$ from B to A 4) $2\mu\text{C}$ from A to B
46. Three condensers of same capacity connected in series has effective capacity 2mF . If they are connected in parallel and charged using a battery of emf 12V , the total energy stored in the combination is
- 1) 1296 mJ 2) 648 mJ 3) 162 mJ 4) 48 mJ
47. The capacity of a condenser is $50\mu\text{F}$. It has charged to a p.d. of 100 volts by a battery. The distance between the plates is decreased to half by maintaining the battery connection then energy stored in condenser is
- 1) $50 \times 10^{-2}\text{ J}$ 2) $250 \times 10^{-2}\text{ J}$ 3) $120 \times 10^{-2}\text{ J}$ 4) $20 \times 10^{-2}\text{ J}$

48. Two identical capacitors are in parallel and are connected to a battery of emf 3000V, store 1120J of energy. When they are connected in series to the same battery, the energy stored in each capacitor is
 1) 378 J 2) 140 J 3) 530 J 4) 70 J
49. A capacitor of 8 micro farad is charged to a potential of 1000V. The energy stored in the capacitor is
 1) 4J 2) 14J 3) 12J 4) 10J
50. A $4\mu\text{F}$ capacitor is charged by a 200 V battery. It is then disconnected from the supply and is connected to another uncharged $2\mu\text{F}$ capacitor. During the process loss of energy (in J) is
 1) Zero 2) 5.33×10^{-2} 3) 4×10^{-2} 4) 2.67×10^{-2}
51. A body of capacity $6\mu\text{F}$ is charged to 20V and another body of capacity $4\mu\text{F}$ is charged to 10 V. When they are connected the energy lost by the system is
 1) 0.78mJ 2) 0.46mJ 3) 0.12mJ 4) 0.25 mJ
52. A variable capacitor is kept connected to a 10V battery. If the capacitance of the capacitor is changed from $7\mu\text{F}$ to $3\mu\text{F}$, the work done is
 1) 2×10^{-4} J 2) 4×10^{-4} J 3) 6×10^{-4} J 4) 8×10^{-4} J
53. One plate of a parallel plate capacitor is connected to a spring as shown as shown in the figure. The area of each plate of the capacitor is A and the distance between the plates is d when the battery is not connected and the spring is unstretched. After connecting the battery, in the steady state, the distance between the plates is 0.75d, then the force constant of the spring is (Eamcet-2018)



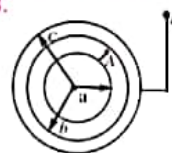
- 1) $\frac{3\epsilon_0 V^2 A}{8 d^3}$ 2) $\frac{8\epsilon_0 V^2 A}{3 d^3}$ 3) $\frac{9\epsilon_0 V^2 A}{32 d^3}$ 4) $\frac{32\epsilon_0 V^2 A}{9 d^3}$
54. Two parallel plate capacitors are arranged perpendicular to the common axis. The separation d between the capacitors is much larger than the separation l between their plates and than their size. The capacitors are charged to q_1 and q_2 respectively (figure). Find the force of interaction between the capacitors.



- 1) $\frac{1}{2} \frac{q_1 q_2}{\pi \epsilon_0 d^4} l^2$ 2) $\frac{2}{3} \frac{q_1 q_2}{\pi \epsilon_0 d^4} l^2$ 3) $\frac{3}{2} \frac{q_1 q_2}{\pi \epsilon_0 d^4} l^2$ 4) $\frac{1}{3} \frac{q_1 q_2}{\pi \epsilon_0 d^4} l^2$

55. Consider an assembly of three conducting concentric spherical shells of radii a, b and c as shown in figure. Find the capacitance of the assembly between the points A and B.

- 1) $\frac{4\pi\epsilon_0 ac}{c+a}$ 2) $\frac{\pi\epsilon_0 ac}{c+a}$
3) $\frac{\pi\epsilon_0 ac}{c-a}$ 4) $\frac{4\pi\epsilon_0 ac}{c-a}$



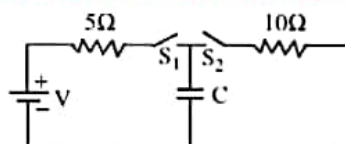
56. Three long concentric conducting cylindrical shells have radii $R, 2R$ and $2\sqrt{2}R$. Inner and outer shells are connected to each other. The capacitance across middle and inner shells per unit length is:

- 1) $\frac{1}{3} \frac{\epsilon_0}{\ln 2}$ 2) $\frac{6\pi\epsilon_0}{\ln 2}$ 3) $\frac{\pi\epsilon_0}{2\ln 2}$ 4) None

57. A parallel plate capacitor at of capacity $100\mu\text{F}$ is charged by a battery at 50 volts. The battery remains connected and if the plates of the capacitor are separated so that the distance between them is halved the original distance, the additional energy gives by the battery to the capacitor in Joule is

- 1) $12 \times 10^{-3} \text{ J}$ 2) $1.2 \times 10^{-3} \text{ J}$ 3) $125 \times 10^{-3} \text{ J}$ 4) $12.5 \times 10^{-3} \text{ J}$

58. In the adjoining diagram, the condenser C will be fully charged to potential V if



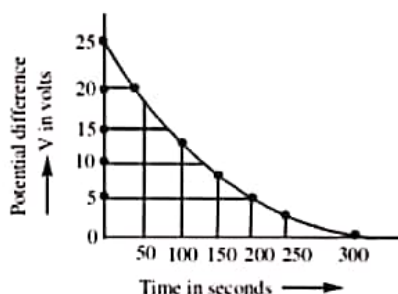
- 1) S_1 and S_2 both are open 2) S_1 and S_2 both are closed
3) S_1 is closed and S_2 is open 4) S_1 is open and S_2 is closed

Previous Mains Problems :

59. A uniform electric field $\vec{E}$ exists between the plates of a charged condenser. A charged particle enters the space between the plates and perpendicular to $\vec{E}$. The path of the particle between the plates is a [Online April 9, 2013]

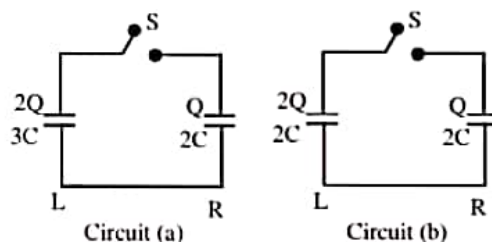
- 1) straight line 2) hyperbola 3) parabola 4) circle

60. The figure shows an experimental plot discharging of a capacitor in an RC circuit. The time constant τ of this circuit lies between [2012]



- 1) 150 sec and 200 sec 2) 0 sec and 50 sec 3) 50 sec and 100 sec 4) 100 sec and 150 sec

61. The capacitor of an oscillatory circuit is enclosed in a container. When the container is evacuated, the resonance frequency of the circuit is 10 kHz. When the container is filled with gas, the resonance frequency changes by 50 Hz. The dielectric constant of the gas is [Online May 26, 2012]
 1) 1.001 2) 2.001 3) 1.01 4) 3.01
62. This question has Statement 1 and Statement 2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
 Statement 1: It is not possible to make a sphere of capacity 1 farad using a conducting material.
 Statement 2: It is possible for earth as its radius is 6.4×10^6 m. [Online May 26, 2012]
 1) Statement 1 is true, Statement 2 is true, Statement 2 is the correct explanation of Statement 1.
 2) Statement 1 is false, Statement 2 is true.
 3) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation of Statement 1.
 4) Statement 1 is true, Statement 2 is false.
63. A series combination of n_1 capacitors, each of capacity C_1 is charged by source of potential difference 4 V. When another parallel combination of n_2 capacitors each of capacity C_2 is charged by a source of potential difference V, it has the same total energy stored in it as the first combination has. The value of C_2 in terms of C_1 is then [Online May 12, 2012]
 1) $16 \frac{n_2}{n_1} C_1$ 2) $\frac{2C_1}{n_1 n_2}$ 3) $2 \frac{n_2}{n_1} C_1$ 4) $\frac{16C_1}{n_1 n_2}$
64. Two circuits (a) and (b) have charged capacitors of capacitance C, 2C and 3C with open switches. Charges on each of the capacitor are as shown in the figures. On closing the switches [Online May 7, 2012]

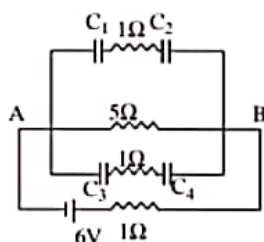


- 1) No charge flows in (a) but charge flows from R to L in (b)
 2) Charges flow from L to R in both (a) and (b)
 3) Charges flow from R to L in (a) and from L to R in (b)
 4) No charge flows in (a) but charge flows from L to R in (b)
65. Let C be the capacitance of a capacitor discharging through a resistor R. Suppose t_1 is the time taken for the energy stored in the capacitor to reduce to half its initial value and t_2 is the time taken for the charge to reduce to one-fourth its initial value. Then the ratio t_1/t_2 will be [2010]
 1) 1 2) $\frac{1}{2}$ 3) $\frac{1}{4}$ 4) 2
66. A parallel plate capacitor with air between the plates has capacitance of 9 pF. The separation between its plates is 'd'. The space between the plates is now filled with two dielectrics. One of the dielectrics has dielectric constant $k_1 = 3$ and thickness $\frac{d}{3}$ while the other one has dielectric constant $k_2 = 6$ and thickness $\frac{2d}{3}$. Capacitance of the capacitor is now [2008]
 1) 1.8 pF 2) 45 pF 3) 40.5 pF 4) 20.25 pF

67. A parallel plate condenser with a dielectric of dielectric constant K between the plates has a capacity C and is charged to a potential V volt. The dielectric slab is slowly removed from between the plates and then reinserted. The net work done by the system in this process is [2007]
- 1) zero 2) $\frac{1}{2}(K-1)CV^2$ 3) $\frac{CV^2(K-1)}{K}$ 4) $(K-1) CV^2$
68. A parallel plate capacitor is made by stacking n equally spaced plates connected alternatively. If the capacitance between any two adjacent plates is ' C ' then the resultant capacitance is [2005]
- 1) $(n+1) C$ 2) $(n-1) C$ 3) nC 4) C
69. A fully charged capacitor has a capacitance ' C '. It is discharged through a small coil of resistance wire embedded in a thermally insulated block of specific heat capacity ' s ' and mass ' m '. If the temperature of the block is raised by ' ΔT ' the potential difference ' V ' across the capacitance is [2005]
- 1) $\frac{mC\Delta T}{s}$ 2) $\sqrt{\frac{2mC\Delta T}{s}}$ 3) $\sqrt{\frac{2ms\Delta T}{C}}$ 4) $\frac{ms\Delta T}{C}$
70. A sheet of aluminium foil of negligible thickness is introduced between the plates of a capacitor. The capacitance of the capacitor [2003]
- 1) decreases 2) remains unchanged 3) becomes infinite 4) increases
71. The work done in placing a charge of 8×10^{-18} coulomb on a condenser of capacity 100 micro-farad is [2003]
- 1) 16×10^{-32} joule 2) 3.1×10^{-26} joule 3) 4×10^{-10} joule 4) 32×10^{-32} joule
72. If there are n capacitors in parallel connected to V volt source, then the energy stored is equal to [2002]
- 1) CV 2) $\frac{1}{2}nCV^2$ 3) CV^2 4) $\frac{1}{2n}CV^2$
73. Capacitance (in F) of a spherical conductor with radius with radius 1 m is [2002]
- 1) 1.1×10^{-10} 2) 10^{-6} 3) 9×10^{-9} 4) 10^{-3}

Numerical Value Type Questions

74. If the area of each plate is S and the successive separations are d , $2d$ and $3d$ then the equivalent capacitance across A and B is _____ times $\frac{\epsilon_0 A}{d}$
75. In the figure shown, each capacitor is $1\mu F$ and the resistors are indicated. The charges Q_1 and Q_4 on the capacitors C_1 and C_4 are respectively under steady condition. Then, $Q_1 =$ _____ μC .

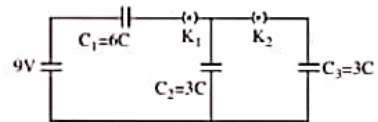


CAPACITORS

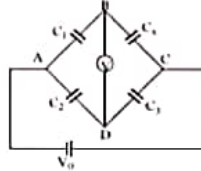
OBJECTIVE PHYSICS IIA & IIB

76. In the circuit shown in figure, Initially K_1 is closed and K_2 is open, let charge on capacitor C_2 is Q_2 . Then K_1 was opened while K_2 kept closed, let charge on capacitor C_2 is Q_2 .

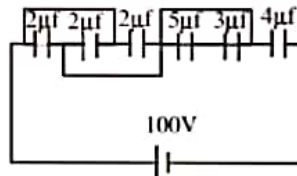
(Take, $C = 1\mu\text{F}$) Then ratio $\frac{Q_2}{Q_2}$ is



77. In capacitor bridge, a voltage V_0 is applied and the variable capacitor C_1 has been adjusted till voltmeter shown zero reading. If $C_1 = 8.9\mu\text{F}$, $C_2 = 18\mu\text{F}$ at balance point, then C_x is ____ μF .



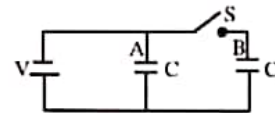
78. In the circuit shown in figure charge stored in the capacitor of capacity $5\mu\text{f}$ is



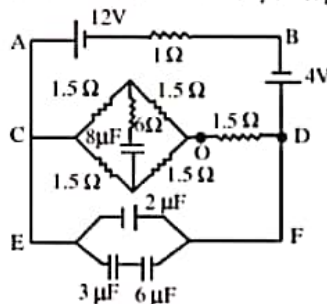
79. The capacity of a parallel plate capacitor formed by the plates of same area A is $0.02\mu\text{F}$ with air as dielectric. Now one plate is replaced by a plate of area $2A$ and dielectric ($k=2$) is introduced between the plates, the capacity is ____ (in μF)

80. The capacity of a parallel plate condenser with air as dielectric is $2\mu\text{F}$. The space between the plates is filled with dielectric slab with $K = 5$. It is charged to a potential of 200V and disconnected from cell. Work done in removing the slab from the condenser completely is ____ J.

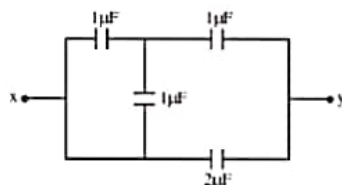
81. Figure shows two identical parallel plate capacitors connected to a battery through a switch S . Initially, the switch is closed so that the capacitors are completely charged. The switch is now opened and the free space between the plates of the capacitors is filled with a dielectric of dielectric constant 3. The ratio of the initial total energy stored in the capacitors to the final total energy stored is



82. In the given circuit, the potential difference across the $6\mu\text{F}$ capacitor in steady state is



83. Four capacitors are connected as shown in figure. Their capacities are indicated in the figure. The effective capacitance between points x and y is (in μF)



KEY SHEET

EXERCISE-I

1) 3	2) 3	3) 1	4) 3	5) 1	6) 2	7) 2	8) 3	9) 4	10) 2
11) 1	12) 2	13) 1	14) 4	15) 2	16) 3	17) 3	18) 2	19) 2	20) 2
21) 4	22) 2	23) 2	24) 1	25) 3	26) 3	27) 4	28) 3	29) 1	30) 1
31) 4	32) 4	33) 3	34) 4	35) 2	36) 3	37) 2	38) 3	39) 2	40) 1
41) 4	42) 1	43) 1	44) 4	45) 3	46) 3	47) 2	48) 3	49) 1	50) 4
51) 4	52) 1	53) 1	54) 2	55) 2	56) 4	57) 1	58) 4	59) 1	60) 2
61) 2	62) 3	63) 4	64) 1	65) 1	66) 4	67) 4	68) 1	69) 1	70) 4
71) 3	72) 3	73) 4	74) 2	75) 3	76) 8	77) 2.51	78) 20	79) 2.33	
80) 0.83	81) 1	82) 8	83) 6	84) 0.02	85) 1.33				

EXERCISE-II

1) 2	2) 2	3) 2	4) 3	5) 1	6) 4	7) 2	8) 4	9) 4	10) 1
11) 3	12) 1	13) 3	14) 2	15) 1	16) 1	17) 4	18) 1	19) 2	20) 3
21) 3	22) 1	23) 3	24) 1	25) 4	26) 3	27) 3	28) 1	29) 1	30) 1
31) 2	32) 2	33) 4	34) 1	35) 1	36) 2	37) 1	38) 2	39) 1	40) 3
41) 4	42) 2	43) 2	44) 4	45) 1	46) 1	47) 1	48) 2	49) 1	50) 4
51) 3	52) 1	53) 4	54) 3	55) 4	56) 2	57) 3	58) 3	59) 3	60) 4
61) 3	62) 4	63) 4	64) 3	65) 3	66) 3	67) 1	68) 2	69) 3	70) 2
71) 4	72) 2	73) 1	74) 0.25	75) 2.5	76) 2	77) 2.4	78) 0	79) 0.04	
80) 0.8	81) 0.6	82) 2	83) 2.67						